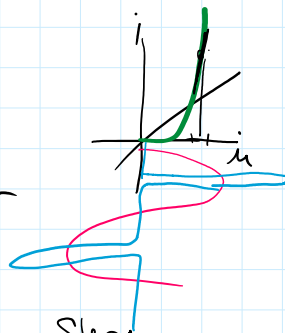


LINEARE ZWEIPOLE

$R \rightarrow u = R \cdot i$ algebraisch

$L, C \rightarrow$ lineare Differentialgleichung



- sinusförmige Spannung $\rightarrow$ sinusförmiger Strom
- Schaltung mit lin. ZP $\rightarrow$ mehrere Quellen sinusförmig $\rightarrow$ sinusförmige Signale
- passive Zweipole $\rightarrow$ Wirkleistung ≥ 0

KOMPLEXER WIDERSTAND UND LEITWERT

$$\underline{Z} = \frac{\underline{U}}{\underline{I}} = \frac{U \angle \varphi_u}{I \angle \varphi_i} = \frac{U}{I} \angle \varphi_u - \varphi_i$$

$$\underline{Z} = Z \angle \varphi_z = R + jX$$

$\swarrow$ komplexer Widerstand (Impedanz)
 $\swarrow$ Scheinwert
 $\swarrow$ Aktivwert
 $\swarrow$ Reaktivwert (Wirkleistung)
 $\swarrow$ Reaktivwert (Blindleistung)

$$\underline{Y} = \underline{I} \angle \varphi_i = \frac{I}{U} \angle \varphi_i - \varphi_u = G + jB$$

$\swarrow$ komplexer Leitwert (Admittanz)
 $\swarrow$ Scheinwert
 $\swarrow$ Aktivwert
 $\swarrow$ Reaktivwert (Wirkleistung)
 $\swarrow$ Reaktivwert (Blindleistung)

$$[\underline{Z}] = \underline{Q}$$

$$Z = \sqrt{R^2 + X^2}$$

$$\varphi = \arctan \frac{X}{R}$$

$$R = Z \cdot \cos \varphi$$

$$X = Z \cdot \sin \varphi$$

$$[\underline{Y}] = \underline{S}$$

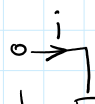
$$Y = \sqrt{G^2 + B^2}$$

$$-\varphi = \arctan \frac{B}{G}$$

$$G = Y \cdot \cos(-\varphi)$$

$$B = Y \cdot \sin(-\varphi)$$

BSP. $\underline{U} = 230V \angle 0^\circ$; $\underline{I} = 1,2A \angle 10^\circ$

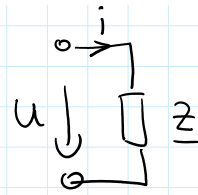


BSP. $\underline{U} = 230V \angle 0^\circ$; $\underline{I} = 1,2A \angle 10^\circ$

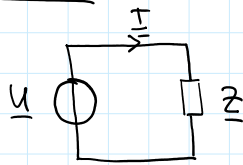
$$\underline{Z} = \frac{\underline{U}}{\underline{I}} = \frac{230V \angle 0^\circ}{1,2A \angle 10^\circ} =$$

$$= 191,6 \Omega \angle -10^\circ = 188,6 \Omega - j 33,28 \Omega$$

$$\underline{Y} = \frac{1}{\underline{Z}} = \frac{1}{Z} \angle -\varphi = 5,2 \text{ mS} \angle 10^\circ = 5,12 \text{ mS} + j 0,9 \text{ mS}$$



BSP. Quelle mit Impedanz



$$\underline{U} = 12V \angle 0^\circ$$

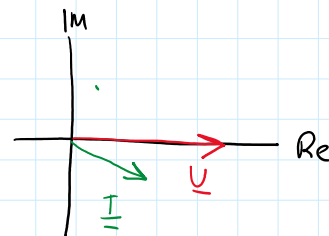
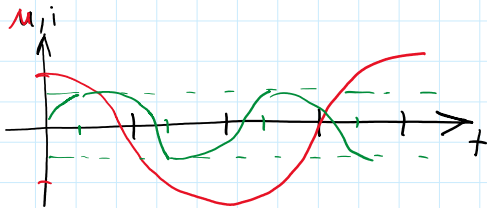
$$f = 10 \text{ Hz}$$

$$\underline{Z} = 4 \Omega \angle 30^\circ$$

$$\underline{I} = \frac{\underline{U}}{\underline{Z}} = \frac{12V \angle 0^\circ}{4 \Omega \angle 30^\circ} = 3A \angle -30^\circ$$

$$\boxed{u(t)} = 12 \cdot \sqrt{2} \cdot \cos(\omega t + 0^\circ) \text{ V}$$

$$\boxed{i(t)} = 3 \cdot \sqrt{2} \text{ A} \cos(\omega t - 30^\circ)$$



DISKUSSION BAUELEMENTE

- Voraussetzung
 - ω = gleich für alle Signale
 - eingeschwungen Zustand

• OHMSCHER WIDERSTAND $u = R \cdot i$

$$\hat{u} \cos(\omega t + \varphi_u) = R \cdot \hat{i} \cos(\omega t + \varphi_i)$$

↓
Koeffizientenvergleich

$$\hat{u} = R \cdot \hat{i} \Rightarrow Z_R = \frac{\hat{u}}{\hat{i}} = R$$

$$\omega t + \varphi_u = \omega t + \varphi_i \Rightarrow \varphi_u = \varphi_i \Rightarrow \varphi_R = 0$$

$$\left. \begin{array}{l} Z_R = R \end{array} \right\} \boxed{\underline{Z}_R = R \angle 0^\circ}$$

• IDEALER INDUKTIVER ZP $u = L \cdot \frac{di}{dt}$

$$\hat{u} \cos(\omega t + \varphi_u) = L \cdot \frac{d}{dt} (\hat{i} \cos(\omega t + \varphi_i)) =$$

$$u = L \cdot \frac{di}{dt}$$

$$\hat{u} \cos(\omega t + \varphi_u) = L \cdot \frac{d}{dt} (\hat{i} \cos(\omega t + \varphi_i)) =$$

$$= L \cdot \hat{i} \omega (-\sin(\omega t + \varphi_i))$$

$$\hat{u} \cos(\omega t + \varphi_u) = \omega L \hat{i} \cos(\omega t + \varphi_i + \frac{\pi}{2})$$

$$\hat{u} = \omega L \hat{i} \Rightarrow Z_L = \omega L$$

$$\omega t + \varphi_u = \omega t + \varphi_i + \frac{\pi}{2} \Rightarrow \varphi_L = \varphi_u - \varphi_i = \frac{\pi}{2}$$

$$Z_L = \omega L \angle \frac{\pi}{2} = j\omega L$$

IDEALER KAPAZITIVER ZP

$$i = C \cdot \frac{du}{dt}$$

$$\hat{i} \cos(\omega t + \varphi_i) = C \cdot \frac{d}{dt} (\hat{u} \cos(\omega t + \varphi_u)) =$$

$$= C \cdot \hat{u} \cdot \omega (-\sin(\omega t + \varphi_u)) =$$

$$= C \cdot \omega \cdot \hat{u} \cos(\omega t + \varphi_u + \frac{\pi}{2})$$

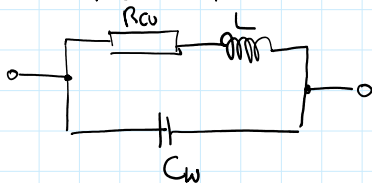
$$\hat{i} = C \cdot \omega \cdot \hat{u} \Rightarrow Z_C = \frac{1}{\omega C}$$

$$\omega t + \varphi_i = \omega t + \varphi_u + \frac{\pi}{2} \Rightarrow \varphi_u - \varphi_i = \varphi_C = -\frac{\pi}{2}$$

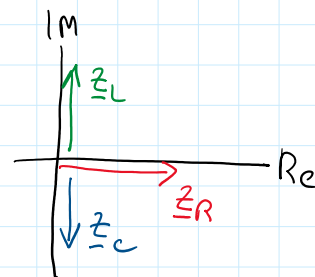
$$Z_C = \frac{1}{\omega C} \angle -\frac{\pi}{2} = -j \frac{1}{\omega C} = \frac{1}{j\omega C}$$

REALE INDUKTIVITÄT

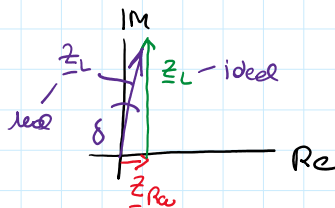
reale Induktivität



R_{cu} ... Widerstand Cu-Wick
 C_w ... Wickkapazität



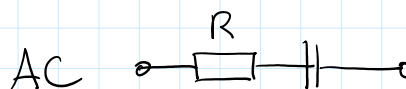
$\omega \downarrow \rightarrow C_w$ vernachlässigen

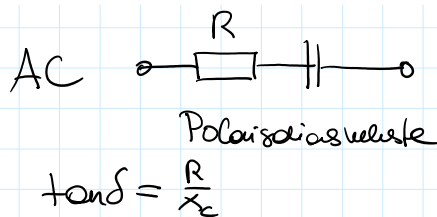
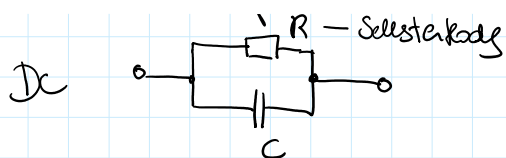


$$\tan \delta = \frac{R}{X_L} \quad \text{Verlustwinkel}$$

$$Q = \frac{1}{\tan \delta} \quad \text{Spulengüte}$$

reale Kapazität





Bauteil	komplexe Leistung $\underline{S}$	$W_{\max}$
R	$R \cdot i^2$ $\emptyset$	$\emptyset$
L	0 $\omega L I^2$	$\frac{L I^2}{2} = L I^2$
C	0 $-\omega C U^2$	$\frac{C U^2}{2} = C U^2$

BSP R, L, C

$$R = 100 \Omega \quad L = 318,31 \text{ mH} \quad C = 31,831 \mu\text{F}$$

$$f_1 = 50 \text{ Hz}, \quad f_2 = 100 \text{ kHz}$$

$$\underline{U} = 230 \text{ V} \angle 0^\circ$$

[R] $\underline{Z}_R = 100 \Omega + j0 \Omega = 100 \Omega \angle 0^\circ \quad (f_1 + f_2)$

$$\underline{I}_R = \frac{\underline{U}}{\underline{Z}_R} = \frac{230 \text{ V} \angle 0^\circ}{100 \Omega \angle 0^\circ} = 2,3 \text{ A} \angle 0^\circ$$

[L] $f_1: \underline{Z}_{L1} = j\omega L = j \cdot 2\pi f \cdot L = j 2\pi 50 \cdot 318 \text{ mH} = j 100 \Omega = 100 \Omega \angle 90^\circ$

$f_2: \underline{Z}_{L2} = \dots = j 2\pi 100 \cdot 10^3 \cdot 318 \text{ mH} = j 200 \text{ k}\Omega = 200 \text{ k}\Omega \angle 90^\circ$

$$\underline{I}_{L1} = \frac{\underline{U}}{\underline{Z}_{L1}} = \frac{230 \text{ V} \angle 0^\circ}{100 \Omega \angle 90^\circ} = 2,3 \text{ A} \angle -90^\circ$$

$$\underline{I}_{L2} = \frac{\underline{U}}{\underline{Z}_{L2}} = \frac{230 \text{ V} \angle 0^\circ}{200 \text{ k}\Omega \angle 90^\circ} = 1,15 \text{ mA} \angle -90^\circ$$

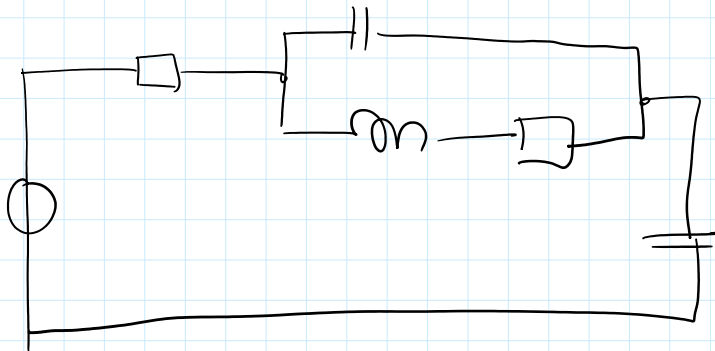
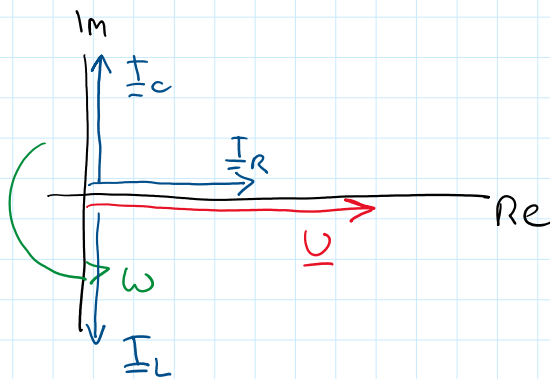
[C] $f_1: \underline{Z}_{C1} = -j \frac{1}{\omega C} = -j \frac{1}{2\pi 50 \cdot 31,8 \cdot 10^{-6}} = -j 100 \Omega = 100 \Omega \angle -90^\circ$

$f_2: \underline{Z}_{C2} = \dots = -j 0,025 \Omega = 0,025 \Omega \angle -90^\circ$

$$\underline{I}_{C1} = \frac{\underline{U}}{\underline{Z}_{C1}} = \frac{230 \text{ V} \angle 0^\circ}{100 \Omega \angle -90^\circ} = 2,3 \text{ A} \angle 90^\circ$$

$$\underline{I} = \frac{\underline{U}}{\underline{Z}} = 4,6 \text{ mA} \angle 90^\circ$$

$$\underline{I}_{c2} = \frac{\underline{U}}{\underline{z}_{c2}} = 4600 \text{ A} \angle 90^\circ$$

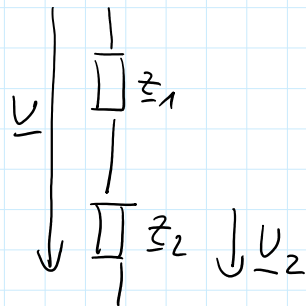


REIHENSCHALTUNG

$$\underline{Z}_e = \sum_{i=1}^n \underline{Z}_i = \sum_{i=1}^n R_i + j \sum_{i=1}^n X_i$$

$$\frac{\underline{U}_2}{\underline{U}} = \frac{\underline{Z}_2}{\underline{Z}_1 + \underline{Z}_2} = \frac{\underline{Z}_2}{\underline{Z}_e}$$

(Spannteiler-Regel)



PARALLELSCHALTUNG

$$\underline{Y}_e = \sum_{i=1}^n \underline{Y}_i = \sum_{i=1}^n G_i + j \sum_{i=1}^n B_i$$

$$\frac{\underline{I}_1}{\underline{I}} = \frac{\underline{Y}_1}{\underline{Y}_e} \quad (\text{Stromteiler-Regel})$$

ENDGÜLTIGE GRÜßE - VERFAHREN

ERSATZQUELLENVERFAHREN

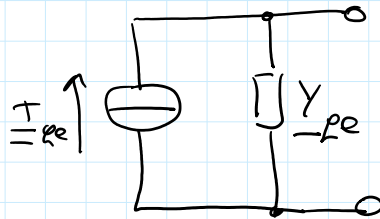
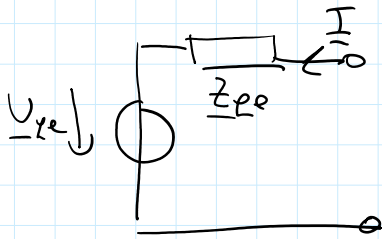
Ersatzquellen $\underline{U}_{qe}$
Ersatzquellenspannung = Spannung des ZP im Leerlauf

Ersatzquellstrom = Strom des ZP im Kurzschluss $\underline{I}_{qe}$

Ersatz-Innenwiderstand = $\underline{U}_{qe} / \underline{I}_{qe}$

$$\underline{U}_{qe} = \underline{U}_0 = \underline{U} \big|_{\underline{I}=0} \quad \underline{I}_{qe} = \underline{I}_k = -\underline{I} \big|_{\underline{U}=0}$$

$$\underline{Z}_{qe} = \frac{1}{\underline{Y}_{qe}} = \frac{\underline{U}_{qe}}{\underline{I}_k}$$



ÜBERLAGERUNGSPRINZIP